calm_rank_tilt

equity strategy | family: equity-volatility | status: research | derived from `calm_weighted`

What this strategy does

Balanced Leverage, with bets shrunk sharply for the jumpiest stocks (ranked against last quarter)

Why it should work. PEAD alpha per dollar is highest in calm stocks; rank-weight toward them without depending on vol levels.

Design notes (from the strategy's source file).

Calm Rank Tilt -- Balanced Leverage, tilted hard toward the calmest stocks by volatility RANK (research).

Calm Weighted sized each event by 1/idiosyncratic-vol (Sharpe 1.30 -> 1.47). The attribution work showed that PEAD alpha per dollar is concentrated in low-volatility names and almost absent in the wildest tercile, and that the 1/vol weight, being linear in the vol LEVEL, only mildly demotes the high-vol tail. Weighting by (1.02 - vol percentile)^2 -- where the percentile is measured against the previous quarter's cross-section within the stock's size quintile -- demotes it hard while staying smooth:

Sharpe 1.30 (Balanced Leverage) -> 1.47 (Calm Weighted) -> 1.81; max drawdown -16.6% -> -11.6%; every era improves split-sample check: exponent picked on 1996-2004 data alone (best k ~ 3) lifts 2005-2013 Sharpe 0.93 -> 1.10, the same as the full-sample best, so the gain is not an artifact of one period.

Lower volatility buys room: `calm_rank_levered` spends it on 1.7x the base unit and a 4x gross cap for ~9% a year at ~5% vol (a market-neutral book clearing the S&P 500's return in this sample). More sizing than that stops paying: the 5%-of-ADV liquidity cap starts to bind and 2008-13 Sharpe decays.

STATUS research: the exponent and sizing were chosen after seeing Calm Weighted's full-sample results (the split-sample check above is the only out-of-sample evidence), and the levered variant leans on small-cap capacity.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	calm rank tilt	market (CRSP VW)
Annualized return	6.01%	9.94%
Annualized volatility	4.04%	19.35%
Sharpe (NAV-pct)	1.47	0.59
Sortino	2.20	

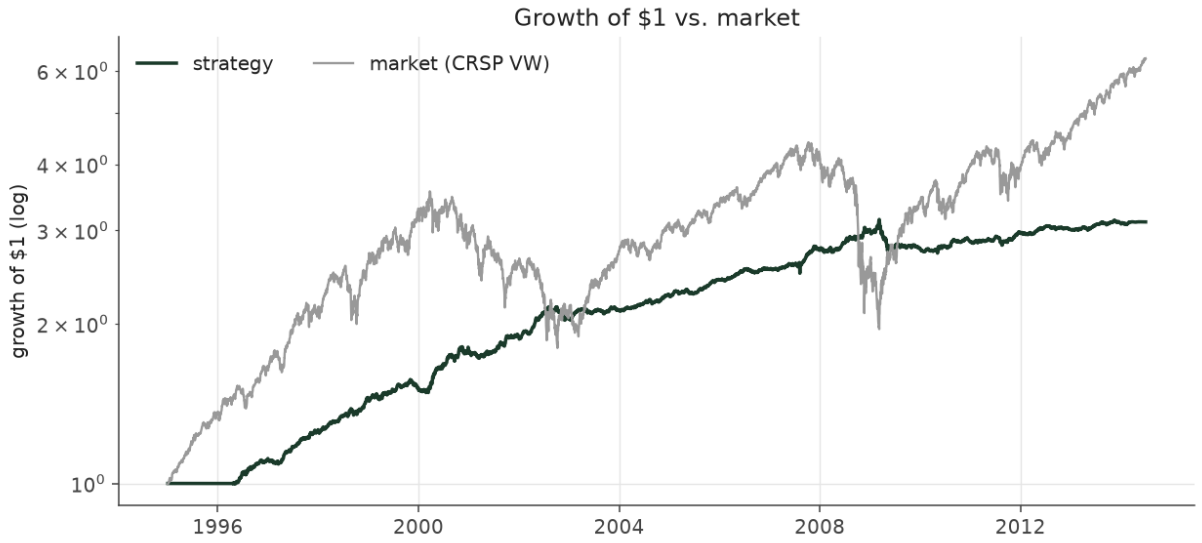
Max drawdown	-13.6%	-55.5%
Calmar	0.44	
Final NAV ($10M start)	$31,136,472	
Worst day	-2.74%	
% positive days	51.6%	
Sharpe 1996-2001	2.31	
Sharpe 2002-2007	1.63	
Sharpe 2008-2013	0.51	

Correlation of daily returns with the market: **-0.09**.

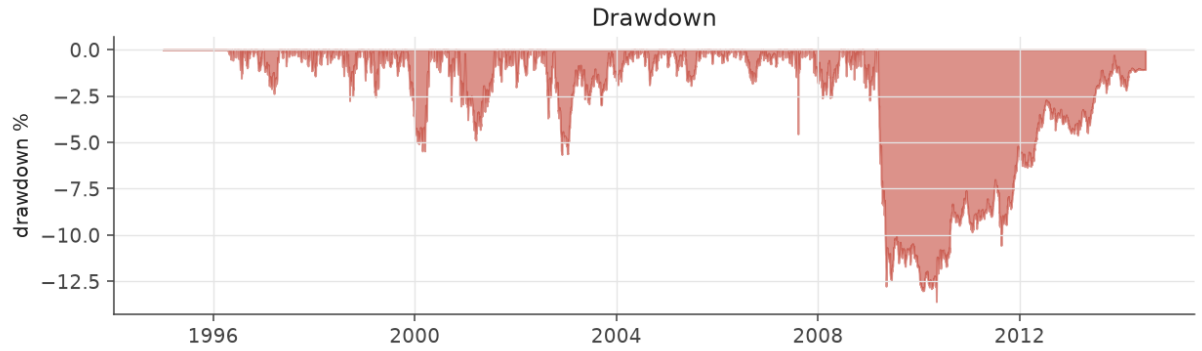
• best Sharpe among equity: this strategy; this one: 1.47

• best return among equity: `market_boost_max` (0.15); this one: 0.0601

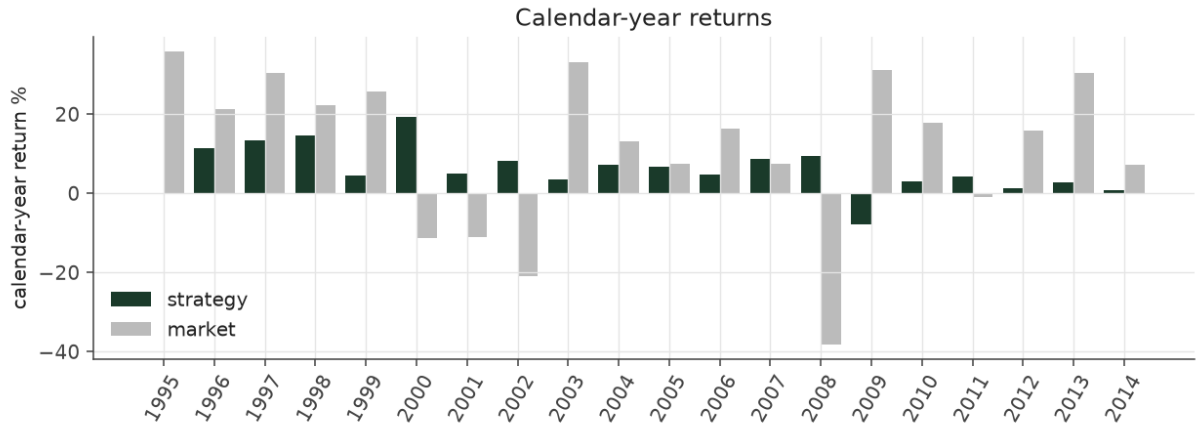
• best Calmar among equity: this strategy; this one: 0.442



NAV (log scale)



Drawdown from running peak

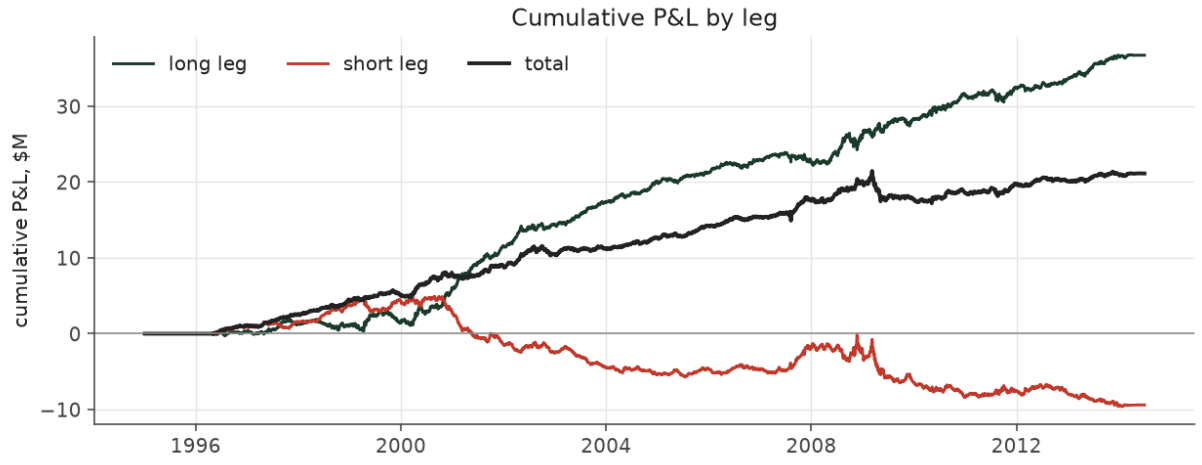


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	36.73	-9.40
P&L after costs ($M)	33.72	-12.58
of which common (average-event) component ($M)	23.99	-24.33

of which excess / PEAD component ($M)	12.74	14.93
excess minus costs ($M)	9.73	11.75
gross return per $ traded (bp)	242.68	-61.14
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	1.35	-0.26

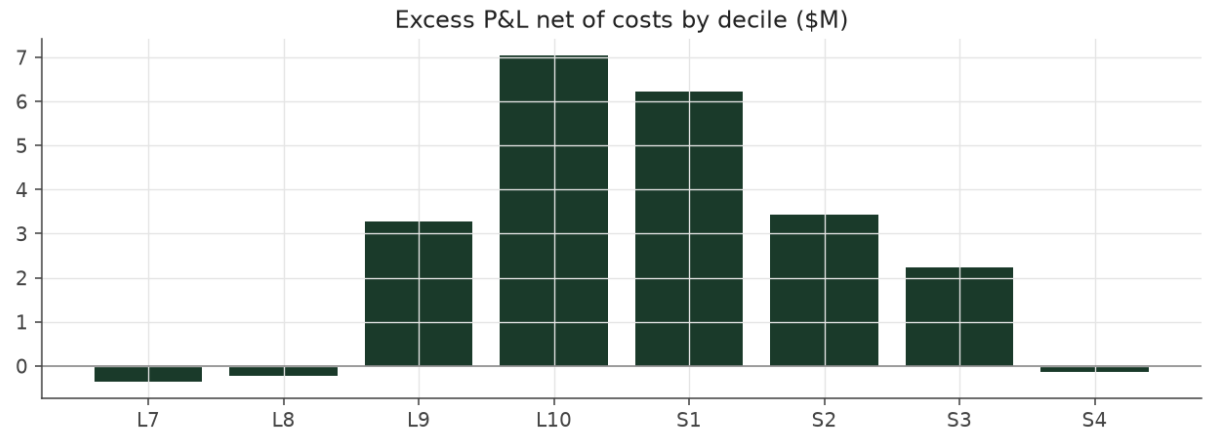
Daily P&L correlation between the two legs: **-0.78**.

• **The short leg loses money in absolute terms** (-9.4M before costs) while the long leg earns +36.7M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 24.0M / 24.3M. Stripping that common component, the surprise signal itself earned +12.7M on the long side and +14.9M on the short side.

• The two legs' daily P&L are -0.78 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.35 long / -0.26 short vs 1.47 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	6,035	104.53	1.45	-0.15	0.21	-0.36	-1.7%	-34.79
long	8	15,853	342.73	5.87	0.47	0.68	-0.22	-1.0%	-6.39
long	9	17,910	462.42	9.63	4.18	0.91	3.27	15.2%	70.70
long	10	19,474	603.92	16.76	8.25	1.21	7.04	32.8%	116.56
short	1	17,681	587.44	-3.43	7.48	1.26	6.21	28.9%	105.79
short	2	18,145	529.86	-5.05	4.52	1.08	3.43	16.0%	64.76
short	3	14,451	375.14	-3.34	2.99	0.75	2.24	10.4%	59.79
short	4	2,253	44.76	-0.76	-0.06	0.08	-0.14	-0.6%	-31.06



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,415	239.05	1.23	2.68	0.94	1.74	8.1%	72.91
2	21,312	542.52	5.65	7.47	1.57	5.90	27.5%	108.76

3	23,965	699.07	5.97	7.47	1.53	5.94	27.7%	85.02
4	26,031	776.73	5.44	6.47	1.27	5.20	24.2%	66.93
5	27,079	793.45	2.85	3.58	0.89	2.69	12.5%	33.91

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,819	192.50	2.26	3.52	0.96	2.56	11.9%	132.96
2	21,720	642.79	4.99	7.90	1.93	5.97	27.8%	92.95
3	24,297	794.77	6.97	8.52	1.59	6.93	32.3%	87.16
4	25,919	739.49	4.43	5.00	1.03	3.97	18.5%	53.66
5	27,043	681.16	2.46	2.70	0.68	2.02	9.4%	29.68

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	39,832	1,082.10	1.89	9.87	2.07	7.80	36.3%	72.05
41-60d	26,568	773.40	4.74	7.77	1.71	6.06	28.2%	78.32
61-90d	36,286	967.96	10.05	6.16	1.78	4.38	20.4%	45.27
>90d	9,116	227.35	4.46	3.87	0.63	3.24	15.1%	142.55

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,241	608.27	9.26	10.60	1.22	9.39	43.7%	154.33
2002-07	36,999	1,061.00	8.16	9.99	2.11	7.88	36.7%	74.24
2008-13	42,562	1,381.53	3.72	7.08	2.87	4.21	19.6%	30.48

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	20,214	298.84	2.27	2.80	0.65	2.15	10.0%	72.01
Chems	2,495	204.06	1.24	1.57	0.36	1.21	5.7%	59.49
Durbl	2,774	217.97	1.31	1.80	0.40	1.40	6.5%	64.31
Enrgy	6,062	242.48	2.38	2.90	0.46	2.44	11.4%	100.63
Hlth	10,828	288.60	1.24	1.81	0.60	1.21	5.6%	41.87
Manuf	11,469	276.55	1.52	2.24	0.58	1.66	7.7%	59.98
Money	16,275	267.89	2.41	3.10	0.61	2.49	11.6%	92.85
NoDur	4,828	247.96	1.54	1.96	0.47	1.49	6.9%	60.02
Other	16,968	289.39	2.61	3.33	0.66	2.68	12.5%	92.47
Shops	11,324	284.11	2.35	2.90	0.60	2.30	10.7%	80.90

Telcm	3,887	233.43	1.78	2.21	0.43	1.79	8.3%	76.49
Utils	4,678	199.52	0.49	1.03	0.37	0.67	3.1%	33.34

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	35,928	249.00	1.63	2.32	0.47	1.85	8.6%	74.44
low vol	38,443	1,738.39	12.00	16.30	3.64	12.67	59.0%	72.86
mid vol	35,681	1,009.55	6.70	8.19	1.98	6.21	28.9%	61.52

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	6.01%	-0.00%
0.5%	1.5	5.73%	0.27%
1.0%	3.1	5.44%	0.56%
2.0%	6.1	4.82%	1.18%
5.0%	15.3	2.39%	3.62%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.